

Predicting Loss Curves of LLM Pretraining

Cosine-to-WSD transfer with reproduced scaling-law baselines and a progress-aware correction

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Repository: <https://github.com/Xiaoxiao-2002/TDLT2026-final-project.git>

1.492%

WSD MAPE after our progress-aware
Tissue correction

Project Requirements Covered

Reproduction

Implement Tissue / momentum law and Multi-Power Law under the same cosine-to-WSD setting.

Diagnosis

Use residuals and schedule-wise prediction plots to locate baseline failure modes.

Method

Propose a lightweight correction fitted on cosine and tested on WSD and 8-1-1.

3

learning-rate schedules: cosine, WSD, 8-1-1

2

reproduced baselines: Tissue and MPL

1

interpretable progress-aware variant

Motivation

Pretraining loss curves are often the earliest reliable signal of whether a run is healthy.

- Full LLM pretraining runs are expensive.
- Learning-rate schedules change the entire optimization trajectory.
- A transferable loss law lets us estimate a new schedule before running a full pretraining experiment.

Central question

Can a loss law fitted only on a cosine schedule predict the loss curve under WSD?

The project is about cross-schedule transfer, not only curve fitting.

Data and Evaluation Protocol

Fit

Course pkl curve with cosine learning-rate schedule.



Predict

WSD is the required target schedule; 8-1-1 is an extra stress test.



Evaluate

Use log-space R^2 , MAPE, MAE, RMSE, PredE, and residual plots.

input: learning-rate history

output: predicted loss curve

train: cosine

test: WSD + 8-1-1

Related Methods We Reproduce

Tissue / Momentum Law

Compact and interpretable law based on learning-rate accumulated features.

Strength: stable fitting and readable parameters.

Risk: too rigid for the early-stage loss drop.

Multi-Power Law

More expressive: seven-parameter model for loss prediction across schedules.

Strength: richer functional form.

Risk: harder to fit and less directly interpretable.

Our design choice: keep Tissue's interpretability, then add the correction suggested by its residual pattern.

Baseline 1: Tissue / Momentum Law

$$L(t) = L_0 + AS_1(t)^{-\alpha} - CS_2(t)$$

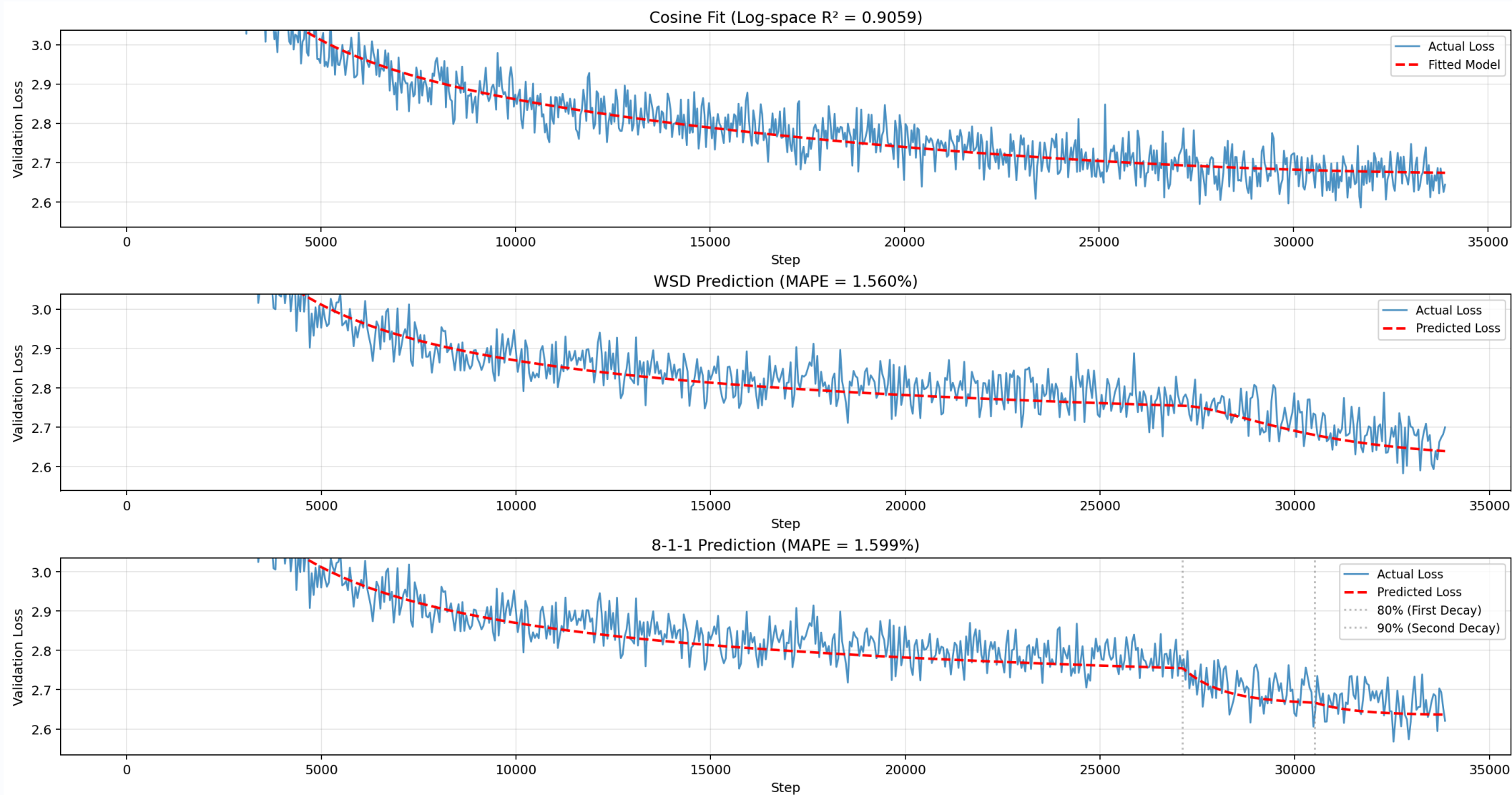
- Fit parameters on cosine.
- Predict WSD and 8-1-1 without refitting.
- Main metric for comparison: WSD MAPE.

1.560%

WSD MAPE for plain Tissue baseline

Observed issue: Tissue fits the general trend, but its residuals reveal biased predictions.

Tissue Baseline: Fit and Transfer



Cosine fit, WSD transfer, and 8-1-1 transfer from the saved Tissue baseline run.

Baseline 2: Multi-Power Law

$$L(t) = L_0 + AS_1(t)^{-\alpha} + B \cdot \text{drop}(t)$$

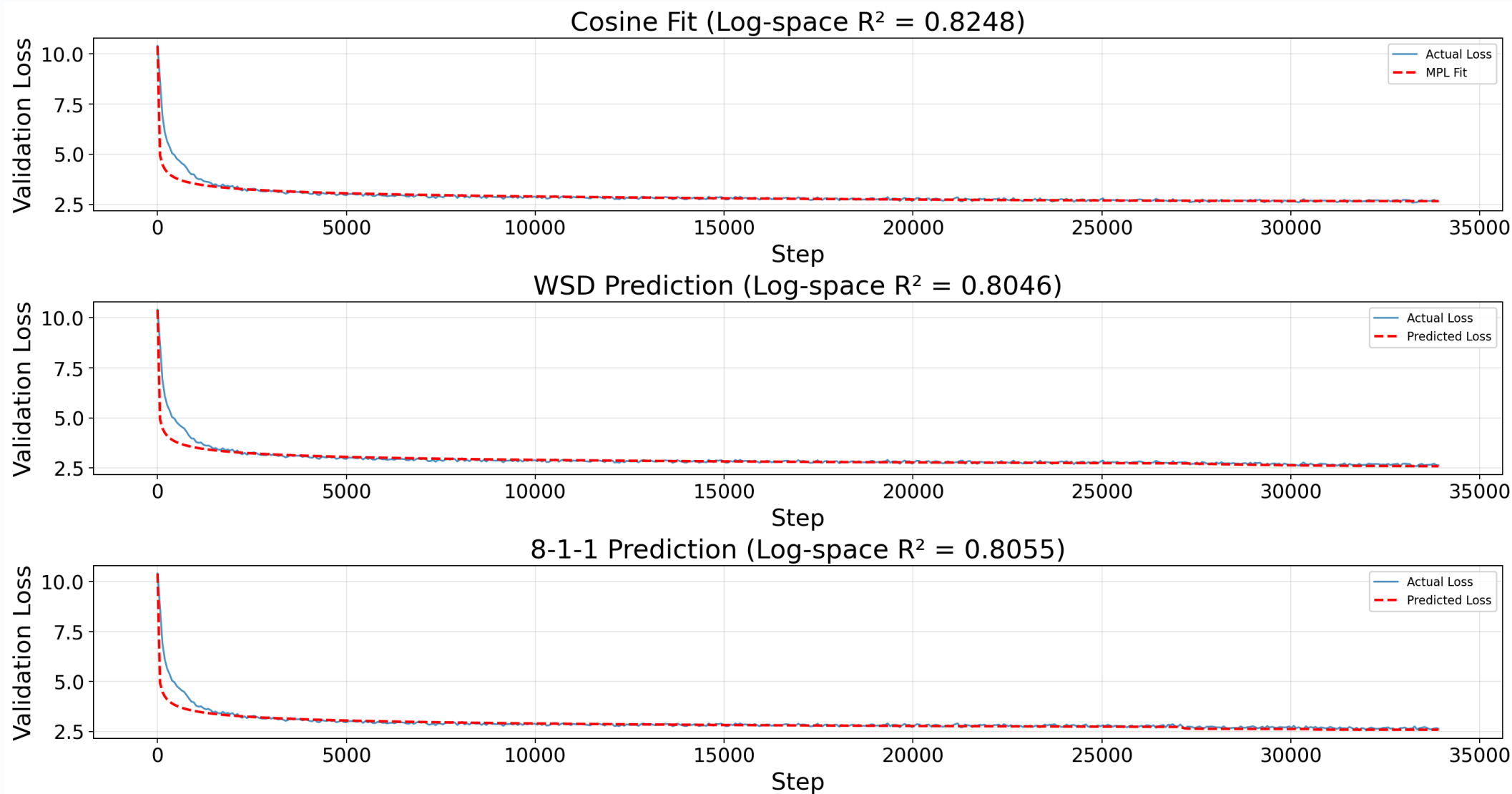
$$\text{drop}(t) = \sum_{i=1}^t (\eta_i - \eta_{i-1}) \left[1 - \left(1 + c\eta_i^{-\gamma} (S_1(t) - S_1(i)) \right)^{-\beta} \right].$$

- Full pkl-based reproduction in `reproduce_2.py`.
- Downsamples the dense pkl curve before fitting, similar to a sparse-checkpoint setup.
- Serves as the more expressive reproduced contrast baseline.

0.805

average test log-space R^2 for MPL

Multi-Power Law: Fit and Transfer

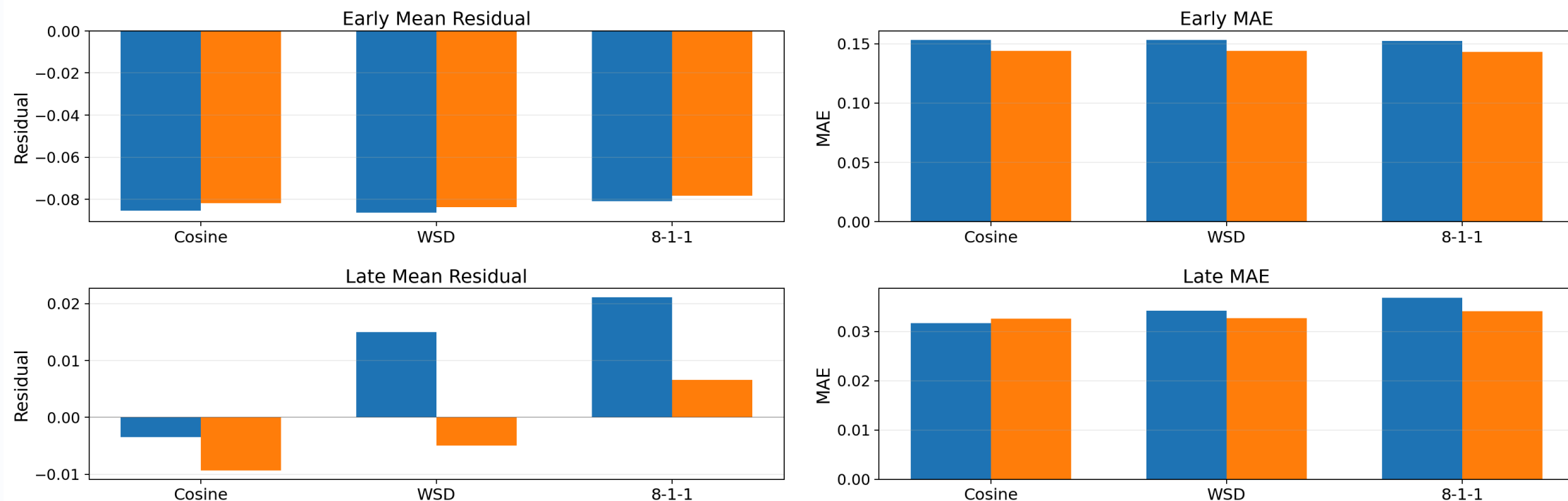


MPL is more expressive, but the pkl-based reproduction is not stronger than the corrected Tissue variant on the chosen metric.

Limitation Analysis: Where Tissue Fails

Residual Pattern Diagnosis for Tissue Baseline vs Improved Variant

Blue: Tissue baseline | Orange: Tissue + progress correction



Negative residuals mean actual loss is lower than predicted loss; the diagnostic shows where the correction moves bias closer to zero.

Our Method: Progress-Aware Tissue Correction

$$L(t) = L_0 + AS_1(t)^{-\alpha} - CS_2(t) - De^{-\tau p(t)}$$

What changes?

Add a decaying correction term over normalized progress $p(t)$, where $p(t)$ ranges from 0 to 1.

Why this term?

It is strongest early, then vanishes, matching the residual pattern found in the baseline.

Contribution: a small, interpretable schedule-transfer correction instead of a large black-box residual model.

Main Quantitative Results

Method	Fit metric	Cross-schedule metric	Role
Tissue baseline	cosine log-space $R^2 = 0.905891$	WSD MAPE = 1.5601% 8-1-1 MAPE = 1.5994%	reproduced baseline
Tissue + progress correction	cosine log-space $R^2 = 0.912989$	WSD MAPE = 1.4924% 8-1-1 MAPE = 1.4925%	our method
Multi-Power Law	train log-space $R^2 = 0.824753$	test log-space $R^2 = 0.805011$	reproduced contrast

-4.3%

relative WSD MAPE reduction vs Tissue baseline

-6.7%

relative 8-1-1 MAPE reduction vs Tissue baseline

simple

only two added parameters: D and τ

Reproducibility and Code Map

reproduce_1.py

Tissue baseline, our progress-aware variant, residual analysis, and diagnostic figure.

experiments/

Saved configs, fitted parameters, predictions, residuals, and metrics for each run.

reproduce_2.py

Multi-Power Law reproduction with pkl loading, sampled checkpoints, metrics, and plot export.

figures/

Main visual evidence used in the final slides and report.

Division of Labor

Tianze Lu

Main code implementation,
Tissue baseline reproduction,
improved method experiments.

Ruoyu Wang

Final slide design, result
organization and polishing.

Feiyue Ye

Literature notes, result checking,
and final formatting support.

Source code, saved experiment artifacts, figures, report notes, and final slides are all included in the repository.

Conclusions and Next Steps

- Cross-schedule loss prediction is feasible, but schedule-sensitive.
- Tissue is interpretable but misses the early-stage loss drop.
- MPL is a useful reproduced contrast baseline.
- Our progress-aware Tissue variant directly targets the observed failure mode.

Final takeaway

Residual diagnosis can guide small, interpretable improvements to scaling-law transfer.

Next: stronger schedule encoding, phase-wise fitting, or regularized residual modeling.